

AICines | Product Analytics

SHORT ANALYTICAL CASE STUDY

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Analysis period: January 2026 – August 2026

Stack: PostgreSQL • SQL • Power Query • Power BI • DAX

15K	10K	60K	132.26M	\$1.31M
USERS	CREATORS	CONTENT	VIEWS	REVENUE

DATA DISCLOSURE

AICines internal operational data was not publicly available. For this portfolio exercise, a controlled synthetic dataset was designed from publicly observable AICines product characteristics and common product-analytics structures. It is not AICines internal data, and the modeled figures do not represent actual AICines business performance.

The purpose is to demonstrate how I would approach product analytics: translate product characteristics into measurable questions, structure and validate data, build reusable metrics, develop an executive dashboard and turn findings into actionable recommendations.

EXECUTIVE SNAPSHOT

- **Five analytical lenses:** user growth & acquisition, creator activity & rewards, content performance & engagement, monetization, and product activity.
- **Four Power BI pages:** Executive Overview, Creator Analytics, Content & Engagement, and Monetization.
- **399,243 modeled records** across calendar, user, creator, content, event, transaction, reward and performance tables.

ANALYTICAL APPROACH

The project uses PostgreSQL as the analytical data layer and Power BI as the reporting layer. A dimensional structure separates descriptive dimensions from business-process facts, while SQL views and DAX measures provide reusable analytical logic.

Data model

- **Dimensions:** dim_date, dim_users, dim_creators, dim_content.
- **Facts:** fact_user_events, fact_transactions, fact_creator_rewards, fact_content_performance.

Validation & reliability

- Checked row counts, date ranges, event coverage, user/creator coverage and financial aggregations.
- Validated creator rewards and content-performance totals.
- Tested Power BI relationships, filter paths and DAX measures.
- Removed blank categories and ensured chronological month sorting.

Why no significant errors occurred: the synthetic dataset was generated under controlled conditions, allowing keys, categories, dates and relationships to remain internally consistent. SQL and Power BI validation then confirmed the model. In production, continuous monitoring would still be required for duplicates, missing events, bots, anomalies, instrumentation gaps, late-arriving data and schema changes.

KEY FINDINGS

All findings below are synthetic-model findings intended to demonstrate analytical interpretation.

Platform scale

- 15,000 modeled users, 10,000 modeled creators and 60,000 modeled content records.
- The content-performance snapshot contains 132.26M modeled views.

Engagement

- Modeled overall engagement rate is **8.43%**, based on likes, comments, recreates and shares relative to views.
- Content types have relatively close engagement-rate values, so volume should be considered alongside deeper engagement measures.

Creator ecosystem

- **75.8%** modeled Creator Reward Coverage.
- Emerging creators form the largest modeled tier, followed by Growing, Established and Top Creator.

Monetization

- **\$1.31M** modeled revenue across **18K transactions**.
- **10,514** modeled paying users; Average Transaction Value is **\$72.77** and Revenue per Paying User is **\$124.59**.

DECISION-ORIENTED INSIGHTS & RECOMMENDATIONS

A key analytical decision was to respect the limitations of the available data. The content-performance table contains a single 31 August 2026 snapshot, so it was used for cross-sectional comparisons such as content type and genre—not to manufacture a January–August views trend. Dated transaction and event data were used for time-series analysis.

Recommendations

- **Strengthen creator conversion & retention:** track first creation → publication, identify drop-off points and compare retention/output by creator tier.
- **Improve content discovery:** evaluate views alongside engagement and watch time; identify content characteristics linked to deeper consumption.
- **Improve monetization analytics:** track revenue with paying-user growth and compare acquisition channels by both volume and revenue contribution.
- **Evaluate the credit economy:** connect credit purchases with creation and interaction behaviour, then examine free-to-paid conversion.
- **Build cohort, funnel & experimentation analytics:** add retention cohorts, acquisition → creation → publication → engagement → monetization funnels, A/B testing, churn, reactivation and lifetime-value metrics.

HOW I WOULD EXTEND THIS AT AICINES

- **Retention:** user and creator cohorts to measure sustained product value.
- **Activation:** identify behaviours associated with successful onboarding.
- **Content:** analyze drivers of watch time, engagement and repeat consumption.
- **Monetization:** connect behaviour, credit consumption, transaction value and lifetime value.
- **Experimentation:** partner with Product and Growth teams on measurable hypotheses and A/B tests.
- **Stakeholder reporting:** maintain consistent KPI definitions across Product, Growth, Creator and Finance teams.

CONCLUSION

This case study demonstrates an end-to-end product analytics workflow: defining business questions, structuring data, preparing and validating it with SQL, developing reusable DAX measures, building a Power BI reporting experience and translating analytical patterns into recommendations.

The project is intentionally transparent about its data limitations. Public AICines product information establishes the business context, while the operational layer is synthetic for analytical demonstration. With production access, the framework could be extended into retention, cohorts, funnels, experimentation, churn and lifetime-value analysis.

Analytical objective: turn product data into clear evidence that helps teams understand user behaviour, creator health, content performance and monetization — and use those insights to support measurable product decisions.